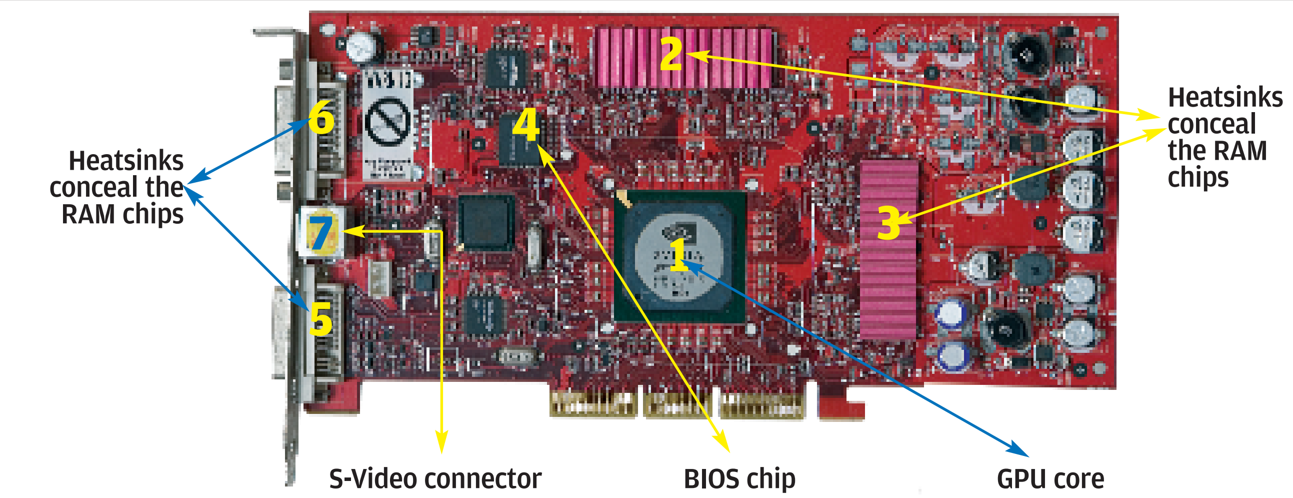

GPU Facts

At the heart of technology in a graphics card is the GPU or Graphics Processing Unit. Think of the GPU as a processor dedicated solely to graphics. Just as a CPU slots into a motherboard, a GPU is affixed on the PCB (Printed Circuit Board) of the graphics card. However, unlike a CPU, a GPU can never be removed from its PCB. Today's GPUs have the ability to number crunch enormous amounts of data and are much more powerful, even than the fastest quad core. NVIDIA's 8800GTX GPU, for example, has 128 stream processors inbuilt and each is capable of handling one thread of data. The constant need for speed has led manufacturers to up transistor counts while technology advances have ensured that the die size of today's GPU core has shrunk. ATI technologies has already made the shift to 55 nm while NVIDIA has gone to a 65 nm process, previously both vendors were using 90 nm and 80 nm fabrication processes. More transistors are being crammed on a smaller GPU die, and the 8800GTX broke all barriers by having 681 million transistors on its 90 nm die—the previous high was 276 million transistors, which was found on ATI's X1950XTX. The brand new 8800GT has crossed the 700 million transistor barrier, and at this rate we could see a billion transistors on a GPU die within a year.

So what increased with this increase in transistor count? For one, the number of Shader units rose steadily, from 4 pixel pipelines (PPs) on the GeForce FX series, to 16 PPs on the GeForce 6800, to 24 PPs on the GeForce 7800GTX, and finally 128 SPs (Stream Processors) on the 8800GTX. Similarly, the colour precision has gone from 16 bits on the GeForce 4 to full 32-bit precision on the 8800GTX, and double precision (64-bit) is on the way. Increases in performance and precision have gone hand in hand with increases in programmability with DX 8.1 giving way to DX 9, and then DX 9.0c (Shader Model 3), and finally DX 10.

The characteristic properties of a GPU mainly depend on two variables:

1. The core speed: this is the rated frequency (in MHz, GHz etc) of the processing unit. A general thumb rule is the faster the core, the faster the GPU. But performance doesn't only depend on the frequency. For example, the core on a GeForce 7900GTX runs at 650 MHz, while the core on a GeForce 8800GTX runs at 575 MHz, in spite of this the 8800GTX is more than twice as fast as the 7900GTX in terms of its processing performance.

2. Pixel and Vertex Shaders: A Vertex Shader basically takes a 3D image made up of vertices and lines and draws a 2D image out of it, which is displayed in the form of pixels on the screen. A Pixel Shader is a refinement of the Vertex Shader and allows the addition of textures to the 2D image on a per pixel basis. With the advent of DX10 both major vendors NVIDIA and ATI have done away with the concept of Pixel and Vertex Shaders completely choosing to design their architectures on a *Unified Shader* model. These Shader units are also called *Stream Processors*. The more the Shader units aboard a GPU the faster it will perform, since it can complete more operations (on pixel and vertex data) in a clock cycle.

So there are two ways to increase performance of a graphics core—increase the clock speed, or increase the count of Shader units. Current limitations of silicon yield mean that increasing clock speeds may not always be possible or even feasible. Increasing the Shader unit count is easier to achieve and the results are startling. Another reason for unifying Shaders would be the move towards making GPUs less graphics specific and more towards general purpose computing engines.

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How Does The Core Work?

Any GPU is tasked with creating an image from a description of a scene, much like an artist. This scene contains a variety of data one of which is geometric primitives (vertices and lines) which are to be viewed in the scene. Also present is the data which contains lighting information for the scene, as well as information about the luminance qualities of the objects in the scene, (objects means the geometric primitive data), i.e. the way that each object reflects light. Finally, information of the viewers' perspective and orientation is also important. This process of image-synthesis has always been viewed as a pipeline process consisting of a number of stages to avoid confusion,

Stage 1: Input

The GPU is designed to work simple vertices and for this purpose every complex shape in a 3D scene is first broken down into triangles—the basic